Tutorial 13 (Relations)

Part I

1. List the ordered pairs in the relation R from $A = \{0, 1, 2, 3, 4\}$ to $B = \{0, 1, 2, 3\}$, where $(a, b) \in R$ if and only if

a) $a = b$.

b) $a + b = 4$.

c) $a > b$.

d) $a \mid b$.

e) $\gcd(a, b) = 1$.

f) $\operatorname{lcm}(a, b) = 2$.

2. a) List all the ordered pairs in the relation $R = \{(a, b) \mid a \text{ divides } b\}$ on the set $\{1, 2, 3, 4, 5, 6\}$.

b) Display this relation graphically, as was done in Example 4.

c) Display this relation in tabular form, as was done in Example 4.

3. For each of these relations on the set $\{1, 2, 3, 4\}$, decide whether it is reflexive, whether it is symmetric, whether it is antisymmetric, and whether it is transitive.

a) $\{(2, 2), (2, 3), (2, 4), (3, 2), (3, 3), (3, 4)\}$

b) $\{(1, 1), (1, 2), (2, 1), (2, 2), (3, 3), (4, 4)\}$

c) $\{(2, 4), (4, 2)\}$

d) $\{(1, 2), (2, 3), (3, 4)\}$

e) $\{(1, 1), (2, 2), (3, 3), (4, 4)\}$

f) $\{(1, 3), (1, 4), (2, 3), (2, 4), (3, 1), (3, 4)\}$

4. Determine whether the relation R on the set of all people is reflexive, symmetric, antisymmetric, and/or transitive, where $(a, b) \in R$ if and only if

a) a is taller than b.

b) a and b were born on the same day.

c) a has the same first name as b.

d) a and b have a common grandparent.

6. Determine whether the relation R on the set of all real numbers is reflexive, symmetric, antisymmetric, and/or transitive, where $(x, y) \in R$ if and only if

a) $x + y = 0$.

b) $x = \pm y$.

c) $x - y$ is a rational number.

d) $x = 2y$.

e) $xy \geq 0$.

f) $xy = 0$.

g) $x = 1$.

h) $x = 1$ or $y = 1$.

7. Determine whether the relation R on the set of all integers is reflexive, symmetric, antisymmetric, and/or transitive, where $(x, y) \in R$ if and only if

a) $x \neq y$.

b) $xy \geq 1$.

c) $x = y + 1$ or $x = y - 1$.

d) $x \equiv y \pmod 7$.

e) x is a multiple of y.

f) x and y are both negative or both non − negative.

g) $x = y^2$.

h) $x \geq y^2$.

10. Give an example of a relation on a set that is

a) both symmetric and antisymmetric.

b) neither symmetric nor antisymmetric.

Exercises 34–37 deal with these relations on the set of real numbers:

$R_1 = \{(a, b) \in \mathbb{R}^2 \mid a > b\}$, the "greater than" relation,

$R_2 = \{(a, b) \in \mathbb{R}^2 \mid a \geq b\}$, the "greater than or equal to" relation,

$R_3 = \{(a, b) \in \mathbb{R}^2 \mid a < b\}$, the "less than" relation,

$R_4 = \{(a, b) \in \mathbb{R}^2 \mid a \leq b\}$, the "less than or equal to" relation,

$R_5 = \{(a, b) \in \mathbb{R}^2 \mid a = b\}$, the "equal to" relation,

$R_6 = \{(a, b) \in \mathbb{R}^2 \mid a \neq b\}$, the "unequal to" relation.

34. Find

a) $R_1 \cup R_3$. b) $R_1 \cup R_5$.

c) $R_2 \cap R_4$. d) $R_3 \cap R_5$.

e) $R_1 - R_2$. f) $R_2 - R_1$.

g) $R_1 \oplus R_3$. *h*) $R_3 \oplus R_4$.

35. Find

a) $R_2 \cup R_4$. *b*) $R_3 \cup R_6$.

c) $R_3 \cap R_6$. *d*) $R_4 \cap R_6$.

e) $R_3 - R_6$. *f*) $R_6 - R_3$.

g) $R_2 \oplus R_6$. *h*) $R_3 \oplus R_5$.

36. Find

a) $R_1 \circ R_1$. *b*) $R_1 \circ R_2$.

c) $R_1 \circ R_3$. *d*) $R_1 \circ R_4$.

e) $R_1 \circ R_5$. *f*) $R_1 \circ R_6$.

g) $R_2 \circ R_3$. *h*) $R_3 \circ R_3$.

37. Find

a) $R_2 \circ R_1$. *b*) $R_2 \circ R_2$.

c) $R_3 \circ R_5$. *d*) $R_4 \circ R_1$.

e) $R_5 \circ R_3$. *f*) $R_3 \circ R_6$.

g) $R_4 \circ R_6$. *h*) $R_6 \circ R_6$.

40. Let R_1 and R_2 be the "divides" and "is a multiple of" relations on the set of all positive integers, respectively. That is, $R_1 = \{(a, b) \mid a \text{ divides } b\}$ and $R_2 = \{(a, b) \mid a \text{ is a multiple of } b\}$. Find

a) $R_1 \cup R_2$. *b*) $R_1 \cap R_2$.

c) $R_1 - R_2$. *d*) $R_2 - R_1$.

e) $R_1 \oplus R_2$.

41. Let R_1 and R_2 be the "congruent modulo 3" and the "congruent modulo 4" relations, respectively, on the set of integers. That is, $R_1 = \{(a, b) \mid a \equiv b \pmod 3\}$ and $R_2 = \{(a, b) \mid a \equiv b \pmod 4\}$. Find

a) $R_1 \cup R_2$. *b*) $R_1 \cap R_2$.

c) $R_1 - R_2$. *d*) $R_2 - R_1$.

e) $R_1 \oplus R_2$.

Part II

1. Represent each of these relations on $\{1,\ 2,\ 3\}$ with a matrix (with the elements of this set listed in increasing order).

a) $\{(1,\ 1),\ (1,\ 2),\ (1,\ 3)\}$

b) $\{(1,\ 2),\ (2,\ 1),\ (2,\ 2),\ (3,\ 3)\}$

c) $\{(1,\ 1),\ (1,\ 2),\ (1,\ 3),\ (2,\ 2),\ (2,\ 3),\ (3,\ 3)\}$

d) $\{(1,\ 3),\ (3,\ 1)\}$

2. Represent each of these relations on $\{1,\ 2,\ 3,\ 4\}$ with a matrix (with the elements of this set listed in increasing order).

a) $\{(1,\ 2),\ (1,\ 3),\ (1,\ 4),\ (2,\ 3),\ (2,\ 4),\ (3,\ 4)\}$

b) $\{(1,\ 1),\ (1,\ 4),\ (2,\ 2),\ (3,\ 3),\ (4,\ 1)\}$

c) $\{(1,\ 2),\ (1,\ 3),\ (1,\ 4),\ (2,\ 1),\ (2,\ 3),\ (2,\ 4),\ (3,\ 1),\ (3,\ 2),\ (3,\ 4),\ (4,\ 1),\ (4,\ 2),\ (4,\ 3)\}$

d) $\{(2,\ 4),\ (3,\ 1),\ (3,\ 2),\ (3,\ 4)\}$

3. List the ordered pairs in the relations on $\{1,\ 2,\ 3\}$ corresponding to these matrices (where the rows and columns correspond to the integers listed in increasing order).

a) $\begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}$ *b*) $\begin{pmatrix} 0 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 0 \end{pmatrix}$ *c*) $\begin{pmatrix} 1 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 1 \end{pmatrix}$

4. List the ordered pairs in the relations on $\{1,\ 2,\ 3,\ 4\}$ corresponding to these matrices (where the rows and columns correspond to the integers listed in increasing order).

a) $\begin{pmatrix} 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 \end{pmatrix}$ *b*) $\begin{pmatrix} 1 & 1 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & 1 \end{pmatrix}$ *c*) $\begin{pmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{pmatrix}$

5. How can the matrix representing a relation R on a set A be used to determine whether the relation is irreflexive?

6. How can the matrix representing a relation R on a set A be used to determine whether the relation is asymmetric?

7. Determine whether the relations represented by the matrices in Exercise 3 are reflexive, irreflexive, symmetric, antisymmetric, and/or transitive.

8. Determine whether the relations represented by the matrices in Exercise 4 are reflexive, irreflexive, symmetric, antisymmetric, and/or transitive.

9. How many nonzero entries does the matrix representing the relation R on $A = \{1, 2, 3, \ldots, 100\}$ consisting of the first 100 positive integers have if R is

a) $\{(a, b) \mid a > b\}$?

b) $\{(a, b) \mid a \neq b\}$?

c) $\{(a, b) \mid a = b + 1\}$?

d) $\{(a, b) \mid a = 1\}$?

e) $\{(a, b) \mid ab = 1\}$?

10. How many nonzero entries does the matrix representing the relation R on $A = \{1, 2, 3, \ldots, 1000\}$ consisting of the first 1000 positive integers have if R is

a) $\{(a, b) \mid a \leq b\}$?

b) $\{(a, b) \mid a = b \pm 1\}$?

c) $\{(a, b) \mid a + b = 1000\}$?

d) $\{(a, b) \mid a + b \leq 1001\}$?

e) $\{(a, b) \mid a \neq 0\}$?

14. Let R_1 and R_2 be relations on a set A represented by the matrices

$$\mathbf{M}_{R1} = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 1 & 1 \\ 1 & 0 & 0 \end{pmatrix} \quad \text{and} \quad \mathbf{M}_{R2} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}$$

Find the matrices that represent

a) $R_1 \cup R_2$.

b) $R_1 \cap R_2$.

c) $R_2 \circ R_1$.

d) $R_1 \circ R_1$.

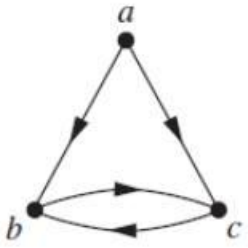
e) $R_1 \oplus R_2$.

19. Draw the directed graphs representing each of the relations from Exercise 2.

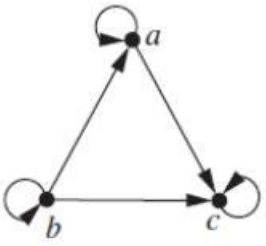
20. Draw the directed graph representing each of the relations from Exercise 3.

In Exercises **23–28** list the ordered pairs in the relations represented by the directed graphs:

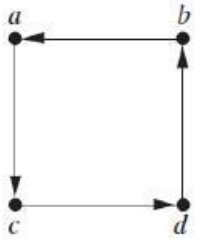
23.



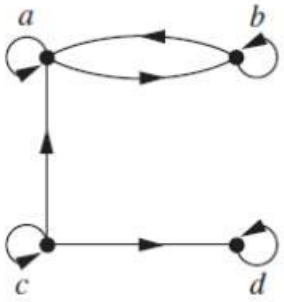
24.



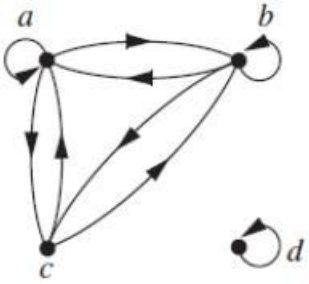
25.



26.



27.



28.





31. Determine whether the relations represented by the directed graphs shown in Exercises 23–28 are reflexive, irreflexive, symmetric, antisymmetric, and/or transitive.

33. Let R be a relation on a set A. Explain how to use the directed graph representing R to obtain the directed graph representing the inverse relation R^{-1}.

36 Given the directed graphs representing two relations, how can the directed graph of the union, intersection, symmetric difference, difference, and composition of these relations be found?

Part III

1. Which of these relations on $\{0, 1, 2, 3\}$ are equivalence relations? Determine the properties of an equivalence relation that the others lack.

a) $\{(0, 0), (1, 1), (2, 2), (3, 3)\}$

b) $\{(0, 0), (0, 2), (2, 0), (2, 2), (2, 3), (3, 2), (3, 3)\}$

c) $\{(0, 0), (1, 1), (1, 2), (2, 1), (2, 2), (3, 3)\}$

d) $\{(0, 0), (1, 1), (1, 3), (2, 2), (2, 3), (3, 1), (3, 2), (3, 3)\}$

e) $\{(0, 0), (0, 1), (0, 2), (1, 0), (1, 1), (1, 2), (2, 0), (2, 2), (3, 3)\}$

3. Which of these relations on the set of all functions from $\mathbf{Z}$ to $\mathbf{Z}$ are equivalence relations? Determine the properties of an equivalence relation that the others lack.

a) $\{(f, g) \mid f(1) = g(1)\}$

b) $\{(f, g) \mid f(0) = g(0)$ *or* $f(1) = g(1)\}$

c) $\{(f, g) \mid f(x) - g(x) = 1$ *for all* $x \in \mathbf{Z}\}$

d) $\{(f, g) \mid$ *for some* $c \in \mathbf{Z}$, *for all* $x \in \mathbf{Z}$, $f(x) - g(x) = c\}$

e) $\{(f, g) \mid f(0) = g(1)$ *and* $f(1) = g(0)\}$

9. Suppose that A is a nonempty set, and f is a function that has A as its domain. Let R be the relation on A consisting of all ordered pairs (x, y) such that $f(x) = f(y)$.

a) Show that R is an equivalence relation on A.

b) What are the equivalence classes of R?

15. Let R be the relation on the set of ordered pairs of positive integers such that $((a, b), (c, d)) \in R$ if and only if $a + d = b + c$. Show that R is an equivalence relation.

16. Let R be the relation on the set of ordered pairs of positive integers such that $((a, b), (c, d)) \in R$ if and only if $ad = bc$. Show that R is an equivalence relation.

17. (Requires calculus)

a) Show that the relation R on the set of all differentiable functions from $\mathbb{R}$ to $\mathbb{R}$ consisting of all pairs (f, g) such that $f'(x) = g'(x)$ for all real numbers x is an equivalence relation.

b) Which functions are in the same equivalence class as the function $f(x) = x^2$?

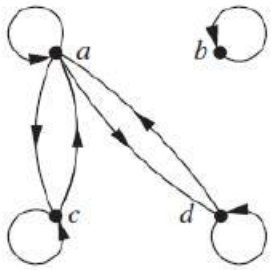
18. (Requires calculus)

a) Let n be a positive integer. Show that the relation R on the set of all polynomials with real-valued coefficients consisting of all pairs (f, g) such that $f^{(n)}(x) = g^{(n)}(x)$ is an equivalence relation. [Here $f^{(n)}(x)$ is the nth derivative of $f(x)$.]

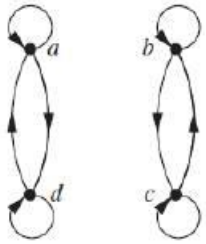
b) Which functions are in the same equivalence class as the function $f(x) = x^4$, where $n = 3$?

In Exercises **21–23** determine whether the relation with the directed graph shown is an equivalence relation.

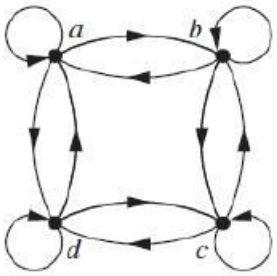
21.



22.



23.



24. Determine whether the relations represented by these zero–one matrices are equivalence relations.

$a)$ $\begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}$

$b)$ $\begin{pmatrix} 1 & 0 & 1 & 0 \\ 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$

$c)$ $\begin{pmatrix} 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$

26. What are the equivalence classes of the equivalence relations in Exercise 1?

27. What are the equivalence classes of the equivalence relations in Exercise 2?

28. What are the equivalence classes of the equivalence relations in Exercise 3?

35. What is the congruence class $[n]_5$ (that is, the equivalence class of n with respect to congruence modulo 5) when n is

$a)$ 2

$b)$ 3

$c)$ 6

$d)$ -3

36. What is the congruence class $[n]_4$ when n is

$a)$ 2

$b)$ 3

$c)$ 6

$d)$ 8

37. Give a description of each of the congruence classes modulo 6.

41. Which of these collections of subsets are partitions of $\{1, 2, 3, 4, 5, 6\}$?

a) $\{1, 2\}, \{2, 3, 4\}, \{4, 5, 6\}$ *b*) $\{1\}, \{2, 3, 6\}, \{4\}, \{5\}$

c) $\{2, 4, 6\}, \{1, 3, 5\}$ *d*) $\{1, 4, 5\}, \{2, 6\}$

42. Which of these collections of subsets are partitions of $\{-3, -2, -1, 0, 1, 2, 3\}$?

a) $\{-3, -1, 1, 3\}, \{-2, 0, 2\}$

b) $\{-3, -2, -1, 0\}, \{0, 1, 2, 3\}$

c) $\{-3, 3\}, \{-2, 2\}, \{-1, 1\}, \{0\}$

d) $\{-3, -2, 2, 3\}, \{-1, 1\}$

44. Which of these collections of subsets are partitions of the set of integers?

a) the set of even integers and the set of odd integers.

b) the set of positive integers and the set of negative integers.

c) the set of integers divisible by 3, the set of integers leaving a remainder of 1 when divided by 3, and the set of integers leaving a remainder of 2 when divided by 3.

d) the set of integers less than -100, the set of integers with absolute value not exceeding 100, and the set of integers greater than 100.

e) the set of integers not divisible by 3, the set of even integers, and the set of integers that leave a remainder of 3 when divided by 6.

46. Which of these are partitions of the set of real numbers?

a) the negative real numbers, $\{0\}$, the positive real numbers

b) the set of irrational numbers, the set of rational numbers

c) the set of intervals $[k, k + 1]$, $k = \ldots, -2, -1, 0, 1, 2, \ldots$

d) the set of intervals $(k, k + 1)$, $k = \ldots, -2, -1, 0, 1, 2, \ldots$

e) the set of intervals $(k, k + 1]$, $k = \ldots, -2, -1, 0, 1, 2, \ldots$

f) the sets $\{x + n \mid n \in Z\}$ for all $x \in [0, 1)$

47. List the ordered pairs in the equivalence relations produced by these partitions of $\{0, 1, 2, 3, 4, 5\}$.

a) $\{0\}, \{1, 2\}, \{3, 4, 5\}$

b) $\{0, 1\}, \{2, 3\}, \{4, 5\}$

c) $\{0, 1, 2\}, \{3, 4, 5\}$

d) $\{0\}, \{1\}, \{2\}, \{3\}, \{4\}, \{5\}$

48. List the ordered pairs in the equivalence relations produced by these partitions of $\{a, b, c, d, e, f, g\}$.

a) $\{a, b\}, \{c, d\}, \{e, f, g\}$

b) $\{a\}, \{b\}, \{c, d\}, \{e, f\}, \{g\}$

c) $\{a, b, c, d\}, \{e, f, g\}$

d) $\{a, c, e, g\}, \{b, d\}, \{f\}$

Part IV

1. Which of these relations on $\{0, 1, 2, 3\}$ are partial orderings? Determine the properties of a partial ordering that the others lack.

a) $\{(0, 0), (1, 1), (2, 2), (3, 3)\}$

b) $\{(0, 0), (1, 1), (2, 0), (2, 2), (2, 3), (3, 2), (3, 3)\}$

c) $\{(0, 0), (1, 1), (1, 2), (2, 2), (3, 3)\}$

d) $\{(0, 0), (1, 1), (1, 2), (1, 3), (2, 2), (2, 3), (3, 3)\}$

e) $\{(0, 0), (0, 1), (0, 2), (1, 0), (1, 1), (1, 2), (2, 0), (2, 2), (3, 3)\}$

2. Which of these relations on $\{0, 1, 2, 3\}$ are partial orderings? Determine the properties of a partial ordering that the others lack.

a) $\{(0, 0), (2, 2), (3, 3)\}$

b) $\{(0, 0), (1, 1), (2, 0), (2, 2), (2, 3), (3, 3)\}$

c) $\{(0, 0), (1, 1), (1, 2), (2, 2), (3, 1), (3, 3)\}$

d) $\{(0, 0), (1, 1), (1, 2), (1, 3), (2, 0), (2, 2), (2, 3), (3, 0), (3, 3)\}$

e) $\{(0, 0), (0, 1), (0, 2), (0, 3), (1, 0), (1, 1), (1, 2), (1, 3), (2, 0), (2, 2), (3, 3)\}$

3. Is (S, R) a poset if S is the set of all people in the world and $(a, b) \in R$, where a and b are people, if

a) a is taller than b?

b) a is not taller than b?

c) $a = b$ or a is an ancestor *of* b?

d) a and b have a common friend?

4. Is (S, R) a poset if S is the set of all people in the world and $(a, b) \in R$, where a and b are people, if

a) a is no shorter than b?

b) a weighs more than b?

c) $a = b$ or a is a descendant *of* b?

d) a and b do not have a common friend?

5. Which of these are posets?

a) $(\mathbf{Z}, =)$ *b*) $(\mathbf{Z}, \neq)$ *c*) $(\mathbf{Z}, \geqslant)$ *d*) $(\mathbf{Z}, \nmid)$

6. Which of these are posets?

a) $(\mathbb{R}, =)$ *b*) $(\mathbb{R}, <)$ *c*) $(\mathbb{R}, \leqslant)$ *d*) $(\mathbb{R}, \neq)$

7. Determine whether the relations represented by these zero–one matrices are partial orders.

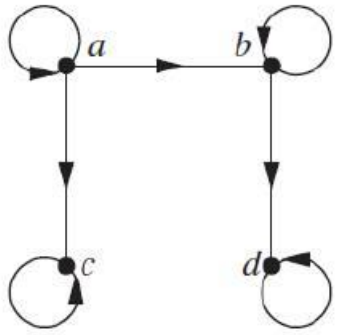
a) $\begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ *b*) $\begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ *c*) $\begin{pmatrix} 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \end{pmatrix}$

8. Determine whether the relations represented by these zero–one matrices are partial orders.

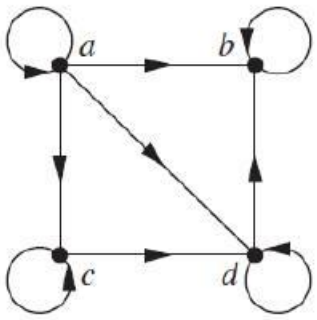
a) $\begin{pmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ *b*) $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}$ *c*) $\begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \end{pmatrix}$

In Exercises **9**–**11** determine whether the relation with the directed graph shown is a partial order.

9.



10.



11.



12. Let (S, R) be a poset. Show that $\left(S, R^{-1}\right)$ is also a poset,where R^{-1} is the inverse of R. The poset $\left(S, R^{-1}\right)$ is called the dual of (S, R).

13. Find the duals of these posets.

a) $(\{0, 1, 2\}, \leq)$ *b*) $(\mathbf{Z}, \geq)$

c) $(P(\mathbf{Z}), \supseteq)$ *d*) $\left(\mathbf{Z}^{+}, |\right)$

14. Which of these pairs of elements are *comparable* in the poset $\left(\mathbf{Z}^+, |\right)$?

a) 5, 15 *b*) 6, 9 *c*) 8, 16 *d*) 7, 7

15. Find two *incomparable* elements in these posets.

a) $(P(\{0, 1, 2\}), \subseteq)$ *b*) $(\{1, 2, 4, 6, 8\}, |)$

16. Let $S = \{1, 2, 3, 4\}$. With respect to the lexicographic order based on the usual "less than" relation,

a) find all pairs in $S \times S$ less than $(2, 3)$.

b) find all pairs in $S \times S$ greater than $(3, 1)$.

c) draw the Hasse diagram of the poset $(S \times S, \preccurlyeq)$.

17. Find the lexicographic ordering of these n-tuples:

a) $(1, 1, 2), (1, 2, 1)$ *b*) $(0, 1, 2, 3), (0, 1, 3, 2)$

c) $(1, 0, 1, 0, 1), (0, 1, 1, 1, 0)$

18. Find the lexicographic ordering of these strings of lowercase English letters:

a) quack, quick, quicksilver, quicksand, quacking

b) open, opener, opera, operand, opened

c) zoo, zero, zoom, zoology, zoological

19. Find the lexicographic ordering of the bit strings 0, 01, 11, 001, 010, 011, 0001, *and* 0101 based on the ordering $0 < 1$.

20. Draw the Hasse diagram for the "greater than or equal to" relation on $\{0, 1, 2, 3, 4, 5\}$.

21. Draw the Hasse diagram for the "less than or equal to" relation on $\{0, 2, 5, 10, 11, 15\}$.

22. Draw the Hasse diagram for divisibility on the set

a) $\{1, 2, 3, 4, 5, 6\}$. *b*) $\{3, 5, 7, 11, 13, 16, 17\}$.

c) $\{2, 3, 5, 10, 11, 15, 25\}$. *d*) $\{1, 3, 9, 27, 81, 243\}$.

23. Draw the Hasse diagram for divisibility on the set

a) $\{1, 2, 3, 4, 5, 6, 7, 8\}$. *b*) $\{1, 2, 3, 5, 7, 11, 13\}$.

c) $\{1, 2, 3, 6, 12, 24, 36, 48\}$. *d*) $\{1, 2, 4, 8, 16, 32, 64\}$.

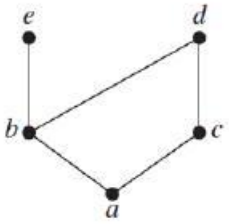
24. Draw the Hasse diagram for inclusion on the set $P(S)$, where $S = \{a, b, c, d\}$.

In Exercises **25–27** list all ordered pairs in the partial ordering with the accompanying Hasse diagram.

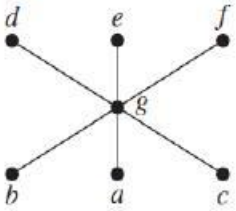
25.



26.



27.



Epilogue

Alhamdulillah, we're done.

Good luck!